

Let ABC be a triangle with $AB = 26$, $BC = 17$, and $AC = 25$. We place B at $(0, 0)$, C at $(17, 0)$, and find A at $(10, 24)$ using coordinate geometry. The feet of the altitudes are $D(10, 0)$ (from A), $E(\frac{9792}{625}, \frac{2856}{625})$ (from B), and $F(\frac{425}{169}, \frac{1020}{169})$ (from C).

Using Menelaus' theorem on $\triangle ABC$ with transversal EFX , we find $BX/XC = 170/119$, leading to $X = D(10, 0)$. The line AX (which is AD) intersects the circumcircle Ω again at $Q(10, -\frac{35}{12})$, the reflection of the orthocenter $H(10, \frac{35}{12})$ over BC .

The nine-point circle (circle DEF) has center at $(\frac{37}{4}, \frac{323}{48})$ and radius $\frac{\sqrt{(3/4)^2 + (323/48)^2}}{2}$. The line HQ (vertical line $x = 10$) intersects this circle at $D(10, 0)$ and $(10, \frac{323}{24})$. Only D lies on segment HQ , so $P = D$.

Calculating distances, $PQ = \frac{35}{12}$ and $PE = ED = \frac{182}{25}$. The ratio $\frac{PQ}{PE} = \frac{35/12}{182/25} = \frac{125}{312}$, where 125 and 312 are coprime. Thus, $a + b = 125 + 312 = 437$.

\boxed{437}